

On-Chain LLM Inference Under Instruction Budgets: An Instruction-Budget Cost Model, Ternary Floor Evidence, and Session Costs

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TL;DR. On the Internet Computer (ICP), large language model (LLM) inference inside a smart-contract canister is bounded not by hardware throughput but by a deterministic per-call instruction budget of about 40 billion metered units. We calibrate a decode cost model, $\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P)$; show that the cost coefficient is software-determined — a 2.9× gap between two forks on identical mainnet weights; produce end-to-end evidence consistent with the ternary floor of the measured decode path ($\alpha_{\text{eff}} \approx 1.00$ published-parameter; $\alpha_{\text{exec}} \approx 1.17$ executed-FLOPs) using a custom TQ2_0 WebAssembly (WASM) SIMD kernel, with byte-identical outputs across a local replica, the Swiss Subnet, and a standard ICP mainnet subnet; and model the per-session IO tax so that the compute budget of stateful multi-call workloads can be planned analytically.

Companion artifact. The companion archive’s `artifact/` directory (MIT code, CC BY 4.0 data; archived at Zenodo DOI 10.5281/zenodo.20607598 and public at GitHub `Simlowker/instruction-bounded-inference-artifact`) contains the calibration registry (49 registry rows plus the on-chain measurement CSVs, spanning local dfx, ICP mainnet, and Swiss Subnet), variance-verified anchors, coherence audits, per-operation profiling, multi-call IO fits, source pins (`artifact/notes/source_pinning.md`), and a source-backed execution-boundary note (`artifact/notes/blockchain_ai_execution_boundaries.md`). The companion reproduction repository `instruction-bounded-inference-artifact` contains the end-to-end local rebuild and re-benchmark procedure for the 2.9× fork gap (`REPRODUCE.md`) and its 2026-05-19 result log (`results/rebench_2026-05-19.md`). Every load-bearing numeric claim below cites an explicit CSV or summary-table path; the full claim-to-evidence mapping lives in `CLAIMS-EVIDENCE-MATRIX.md`.

Abstract

LLM inference on Internet Computer Protocol (ICP) canisters runs under a hard per-call budget of $B = 40 \times 10^9$ instruction-cost units, not wall-clock time. Throughput is therefore bounded by executed instructions, not DRAM bandwidth — the *instruction-bounded* regime. The motivating gap: the same Qwen 2.5 0.5B Q8_0 (8-bit uniform integer) model, on the same ICP mainnet canister, yields 10 tok/call on the onicai baseline and 29 tok/call on our fork. The 2.9× gap is software, not hardware.

(R1) Decode cost model. Decode follows $\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P)$, where α_{eff} is the effective cost coefficient at the published-parameter count $2P$ — a deployment-stack property computable from a model card, not a constant. Across 11 modern decoder models from 8 families, $\alpha_{\text{eff}} = 1.53$ (95% bias-corrected bootstrap CI [1.37, 1.65]; leave-one-architecture-out — LOAO — MAPE 7.7%, model-weighted). Under the executed-FLOPs convention (embedding-table lookups excluded), the physical coefficient α_{exec} clusters at 1.54 (CV 9.3%) for 9 of the 11 models; the two Pythia models — the fit’s only gpt_neox entries, $d \leq 512$ — sit above it ($\alpha_{\text{exec}} 2.15/3.12$), an overhead effect the published-parameter convention masks; at $n = 2$, dimension and family are confounded. Legacy GPT-2 code paths form a separate $\alpha \approx 2.0$ regime under both conventions.

(R2) Matmul dominates. The gap reproduces on mainnet; matmul accounts for 98.7% of decode cost on the profiled Pythia-70M Q8_0 view (98.3% on the Qwen 0.5B Q8_0 SIMD characterization row, §3); per-element unstructured sparsity is provably counter-productive on the measured integer paths (Theorem 1). Swiss Subnet (SSN) [8] deployments stay within 7% of local calibration.

(R3) Floor evidence. A purpose-built TQ2_0 (2-bit ternary in the GGUF binary format used by llama.cpp [10]) WASM SIMD path on purpose-trained TriLM models decodes at the empirical floor of the measured decode path: $\alpha_{\text{eff}} = 1.03$ at 560M ($\alpha_{\text{exec}} = 1.17$ under executed FLOPs), with a coarse integer-token consistency check at 3.9B ($N_{\text{MAX}} = 5$, asymmetric $-17\%/+0.2\%$ resolution, implying $\alpha_{\text{eff}} \leq 1.002$, $\alpha_{\text{exec}} \leq 1.04$) — and byte-identical outputs across a local replica, 13-replica Swiss Subnet consensus, and a standard ICP mainnet subnet. The floor is an observed property of single-token decode on this stack, not a cost-table minimum — isolated ternary kernels bench at $\alpha_{\text{kernel}} = 0.817$ — and the proximity of the published-parameter value to exactly 1.0 is partly a convention coincidence (§2, §5.1). The evidence spans one model family and one format; we do not claim closed saturation.

Three secondary results compose the law into deployment decisions. **(S1)** Uniform Q8_0 stays on the Pareto frontier on SmolLM2-135M and Qwen 0.5B: no mixed-precision variant strictly dominates it, while ΔPPL vs F16 remains tiny. **(S2)** Multi-call sessions pay a per-call IO tax; a hybrid Mamba+Transformer baseline halves the per-token slope. **(S3)** EmbeddingGemma-300M Q4_0 (4-bit uniform integer GGUF) on SSN runs at $\alpha_{\text{embed}} \approx 0.53$, 126 input tok/call — retrieval fits the regime better than long-form decode.

Plain-Language Summary

Running an AI language model on a blockchain is fundamentally different from running it on a GPU. On the Internet Computer (ICP), a smart contract (a “canister”) gets a fixed, deterministic compute budget per call — roughly 40 billion low-level instructions — and the call fails if it exceeds that budget. This paper measures what that constraint means for large language model (LLM) inference in practice. We find a simple scaling law for how many tokens a model can generate per call; show that rewriting the software path (not changing the hardware) can almost triple output; build a ternary-weight WebAssembly (WASM) kernel whose measured cost sits at the floor of the current decode path, with bit-identical outputs across three independent execution environments; and quantify the extra cost of keeping conversation state across calls, so that the compute budget of a multi-call on-chain session can be planned analytically. Whether today’s sub-billion-parameter on-chain models are good enough to make useful agents is a separate, open question (§7.3). The practical recipe: choose the model for quality — today the binding constraint — choose its parameter count and quantization from the instruction law, optimize the matrix-multiplication (matmul) kernel, and treat saved session state as part of the budget.

1 Introduction: On-Chain LLM Inference as an Instruction-Bounded Problem

1.1 From Blockchain AI to a Measurable Problem

We began from a broader blockchain question: can useful AI inference run *natively on-chain*, inside the consensus boundary, rather than through off-chain workers, oracle submissions, proof systems, or precomputed vectors? In the prior-art review that led to this paper, the relevant systems split into two execution boundaries. Some study verifiability, gas costs, or hybrid on/off-chain execution [14, 19, 20]. Others demonstrate native on-chain inference in specific settings, or propose WASM-based blockchain deployments [9, 21], but without a first-order throughput law for decoder execution inside a consensus-metered runtime.

That boundary distinction became the real problem statement of this paper. We are not asking whether “blockchain AI” exists in some broad orchestration sense; we are asking what happens when the model forward pass itself runs under consensus-enforced compute metering. In this study, ICP is the environment where that question becomes directly measurable: deterministic in-consensus WebAssembly execution [5, 6], an explicit per-message compute budget [7], and enough budget to run sub-billion-parameter models end-to-end. That is why the paper centers on ICP while treating the underlying question as broader than ICP itself.

On native hardware, fixing the model, the quantization, and the machine fixes throughput to within a few percent. On ICP, it does not. Qwen 2.5 0.5B Q8_0 generates **10 tokens per update call** on the onicai fork [9] and **29 tokens per update call** on our fork, despite running in the same mainnet canister framework, on the same GGUF weights, under the same 40-billion-instruction message cap.

Once hardware, weights, and quantization are fixed, the remaining explanation is the software path: kernel choice, SIMD integration, dequantization strategy, and the micro-cost of the executed compute graph. Instruction-metered runtimes [5, 6, 7] are therefore governed by different optimization logic than CPUs and GPUs [1, 2, 25]: the dominant cost is executed instructions, not DRAM bandwidth.

1.2 Four questions

This paper answers four questions in order. Each is addressed in one or two sections, and every load-bearing number resolves to a named CSV row in the companion artifact.

1. **What is the regime?** What distinguishes instruction-metered execution from bandwidth-bound inference, and what cost model follows from it? (§1, §2)
2. **What is the law?** Can decode throughput be predicted from total parameter count with a single stack-dependent coefficient? (§3)
3. **Why should the reader believe it?** Does the law survive production deployment (§4), and can a purpose-built kernel reach the floor of its cost model (§5)?
4. **Why does it matter?** How does the per-call law compose into quantization choices and multi-call agentic budgets, and how does the regime extend to non-decode workloads? (§6, §7)

The regime distinction is structural:

	Bandwidth-bound	Instruction-bound
Dominant cost	DRAM bandwidth	Executed instruction count
Hard budget	Wall-clock (soft)	40 B instructions (hard)
Winning quantization	Fewer bits/weight	Fewer instructions per MAC
<code>f32.mul</code> vs <code>i32.mul</code>	Identical	2× per opcode table (scalar; SIMD <code>f32x4/i32x4</code> are priced equal — §4.2)
Main hotspot	Weight loading	Matmul alone (98.3-98.7% across profiled views)

We model the instruction budget rather than the memory ceiling because metering is the runtime’s protocol-level primitive — the constraint that scales with parameter count and prices every call — whereas the wasm-heap and page-access limits that currently bind deployments at sub-1B scales (§6.3, §8) are implementation ceilings of the present runtime build.

The *qualitative* regime distinction — executed instructions, not DRAM bandwidth, as the binding cost — plausibly extends to other discretely-metered runtimes: blockchain canisters [5, 7], fuelled WASM (Wasmtime `set_fuel`, WasmEdge `gas`), and possibly zero-knowledge VMs [23, 24]. The *quantitative* results here (β , the `f32/i32` ratio, α_{eff} , the matmul-dominance share) are calibrated to the ICP opcode cost table and are not claimed to transfer numerically; constraint-counting zkVMs in particular have a different cost structure (proving cost can be super-linear in sequence length), where the linear $2P$ relation need not hold. All measurements in this paper are on the ICP/SSN stack. What transfers is the method, not the constants: on another discretely-metered runtime, re-derive B (the per-invocation budget), re-measure the metering overhead β on a tight-loop microbench, and re-fit α_{eff} from a small model ladder; the regime prediction is that α_{eff} clusters by software path rather than by architecture.

1.3 Contributions and related work

We establish three main results (R1–R3) and three secondary results (S1–S3), all traced to the artifact’s measurement registry (49 rows) and on-chain CSVs. Prior on-chain ML work [14, 19, 20] includes on-chain inference, gas-cost modelling, and verifiability-oriented system design, but does not provide a first-order scaling law for native instruction-metered decoder throughput on a live ICP/WASM canister stack. Concurrent on-chain inference efforts [9, 21] establish that native deployment is possible, but they describe systems and boundaries rather than a parametric law. The closest prior system is opML [28], which runs the LLM forward pass off-chain and re-executes only a single challenged step on-chain inside a fraud-proof VM; its metering is therefore *lazy* — engaged only when a result is disputed (the multi-phase variant additionally verifies phase transitions with on-chain zero-knowledge checks, a hybrid within the lazy family) — whereas the ICP setting meters *every* replica’s full forward pass on *every* update call, eagerly, and our contribution is a throughput cost model rather than a verification protocol. Optimistic-TEE-rollup inference [29] and deterministic-WASM execution [30] target adjacent verifiable-compute boundaries. The practical decision boundary between these designs is the trust root and the latency budget: native in-consensus execution is warranted when the forward pass itself must be a consensus fact — no hardware enclave as trust anchor [29], no challenge window before finality [28], no prover in the loop [24] — and it is dominated by TEE or off-chain-plus-proof designs when model quality or interactive latency is the binding constraint, since the measured envelope here is sub-1B models at tens of tokens per call (§6.2, §7.3). The novelty claimed here is therefore boundary-specific: native decoder execution under consensus metering, measured as a throughput law, not “blockchain AI” in the broad sense. On-chain embedding offerings typically store precomputed vectors or rely on off-chain workers; our encoder case study (§7) runs the full forward pass inside the consensus boundary and is framed as a systems demonstration, not a claim of priority.

2 The Instruction-Bounded Cost Model for Decoder Inference

ICP canisters execute WebAssembly with deterministic instruction metering [5, 6]. Each opcode has an explicit cost per the replica instrumentation cost table [6]; one update call may consume at most $B = 40 \times 10^9$ ICP cost units [7]. Two asymmetries matter for inference. Scalar `f32` arithmetic is priced 2× above scalar `i32` under the ICP opcode cost table (`f32`

2.mul = 2, i32.mul = 1); at the SIMD level, however, the table prices f32x4.mul and i32x4.mul identically (both 2 — instrumentation.rs, cost table inspected at commit eee0fd29765, 2026-05-08). The reconciliation with the observed kernel ratio therefore has two steps: the table-level asymmetry of the measured mul-add kernels is $\approx 4:3$ (from the surrounding instruction mix, not the multiplies), and the metering instrumentation block that dominates a tight inner loop flattens that residual to the observed $1.05\times$ (artifact/data/kernel/matmul_bench.csv). If anything this strengthens §4.2’s conclusion: opcode-mix engineering has even less leverage than a scalar reading of the table suggests. We derive $\beta = 1.844$ for tight SIMD kernels from an empty-loop microbench — one measured quantity (the metered empty-loop cost, Empty_loop_metering row of matmul_bench.csv) combined with one stored accounting estimate (the unmetered per-iteration baseline); β is the metering-overhead multiplier on a tight SIMD inner loop. The full derivation, and the reconciliation between the $2\times$ opcode-table cost and the $1.05\times$ observed SIMD-kernel ratio, is given in artifact/notes/beta_derivation.md.

For decoder-only transformers in the calibrated regime, FLOPs per generated token are well-approximated by $2P$ [16] (with [1, 2] providing the inference-economics and roofline framing). We decompose per-token cost as

$$C_{\text{token}} \approx \alpha_{\text{eff}} \times 2P,$$

and therefore decode throughput as

$$\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P).$$

We distinguish the cost coefficient along two axes. Along the *measurement* axis: α_{kernel} (isolated matmul benches, square matrices, random data), α_{eff} (end-to-end decode throughput; folds in dequantization, dispatch, tiling, metering, and framework overhead; $\alpha_{\text{eff}} > \alpha_{\text{kernel}}$), and α_{embed} (encoder forward passes). Along the *counting-convention* axis: α_{eff} divides by $2 \cdot P_{\text{published}}$ — every number in this paper uses this convention unless suffixed — while α_{exec} divides by *executed* FLOPs only: P_{exec} excludes the input token-embedding table (a per-token row lookup) whenever a separate output head exists, and any learned position embeddings (artifact/notes/alpha_exec_convention.md; recompute in artifact/results/current/scaling_law/alpha_exec.csv). α_{eff} is the deployment predictor — computable from a model card; α_{exec} is the physical quantity — invariant to export-duplicated stored tensors. For tied-embedding models, the majority of the calibration set, the two coincide.

α_{eff} is not a constant of nature. It is a property of the deployment stack: family, quantization, compiler, metering, and implementation. On the current stack, α_{eff} resolves into three distinct clusters:

- **Modern decoder families (FP/INT path):** $\alpha_{\text{eff}} \approx 1.53$ — $\alpha_{\text{exec}} \approx 1.54$ for the nine models outside the small-dimension regime
- **Small-dimension / gpt_neox outliers (n = 2; dimension and family confounded):** masked under α_{eff} ; $\alpha_{\text{exec}} \approx 2.2\text{--}3.1$ (Pythia rows, §3)
- **Legacy GPT-2 code paths:** $\alpha \approx 2.0$ under both conventions (software-penalty regime)
- **Purpose-built TQ2_0 ternary path:** $\alpha_{\text{eff}} \approx 1.00$, $\alpha_{\text{exec}} \approx 1.17$ (empirical decode-path floor)

The floor is load-bearing, so we state precisely what it is — and what it is not. Setting $\alpha_{\text{eff}} = 1$ corresponds to one metered unit per FLOP at the $2P$ count (equivalently, two units per multiply-accumulate, since $2P$ counts multiply and add separately). It is *not* a cost-table minimum: fused SIMD instructions retire several MACs at once, and both the isolated ternary kernel bench (Ternary_DOT_i32x4dot_x4, $\alpha_{\text{kernel}} = 0.817$, artifact/data/kernel/matmul_bench.csv) and the batched encoder path ($\alpha_{\text{embed}} \approx 0.53$, §7.1) measure below 1 on the same cost model. What makes 1 the operative floor for *decode* is structural: single-token matrix-vector decode streams every weight through load + dequantize + MAC exactly once per token, so the per-weight load/dequant instructions cannot be amortized — unlike batched encode, where they are shared across the token batch, and unlike matrix-matrix kernel benches, where operand reuse hides them. On the currently-integrated decode kernels, that per-weight overhead lands the end-to-end ternary coefficient at $\alpha_{\text{exec}} \approx 1.17$ ($\alpha_{\text{eff}} \approx 1.00$ under the published-parameter convention — the roundness of that value is partly a coincidence: TriLM’s untied input table inflates the $2 \cdot P_{\text{published}}$ denominator), and no end-to-end decode deployment on this stack is observed below it under either convention. The claim “the TQ2_0 path reaches the floor” (§5) should therefore be read as measurement evidence that no large remaining margin is visible on that measured decode path at integer-token resolution — an *empirical* floor of the current path, not a bound future kernels cannot cross: the ternary-native DOT $\times 4$ microkernel already measures $\alpha_{\text{kernel}} = 0.817$ on isolated benches (§5.3), and integrating it at model level is the obvious next step.

3 Empirical Decode Scaling Law for Instruction-Bounded LLM Inference

We calibrate the law on local ICP deployments (dfx 0.31.0 — ICP’s canister CLI — fork build 17806d52), batch size 1, binary search on maximum generated tokens before EXCEEDED_INST_LIMIT. The calibration set is scoped to modern decoder families on the current release line; GPT-2 rows are a legacy sensitivity regime; TQ2_0 rows are discussed in §5 as a distinct regime.

Table 1. Representative calibration rows (`artifact/data/models.csv`, `artifact/data/onchain/local.csv`). The first seven rows are the modern-regime calibration examples used to illustrate the main cluster; the final two GPT-2 rows are legacy sensitivity examples shown for contrast and are not part of the modern-only summary statistics reported below. The modern-only fit comprises **eleven models**: these seven rows plus four not shown here for space — Pythia-14M, Mamba-370M, Qwen3-0.6B, and RWKV7-0.4B. Four Transformer anchor points (†) are variance-verified with 3 repetitions each ($CV < 1.05\%$ under the binary-search protocol described above; for transparency, `variance.csv` also retains an earlier same-evening SmolLM2-135M session recording [150, 150, 75] — $CV \approx 35\%$ — traced during revision to a different metric: its values exceed the single-call instruction boundary (94–97 tokens across builds; a single - n 150 call traps IC0522), consistent with cap-limited multi-call totals recorded before the protocol was fixed — the single-call impossibility is proven, the multi-call attribution is inference (`artifact/results/paper_1_5/raw/variance-rootcause-2026-07-09/`); the adopted anchors were re-measured minutes later at [97, 97, 97] under the single-call protocol used throughout, and the reported CIs are conditional on that protocol); a fifth variance-verified point, Mamba-370M, is **retained in the modern fit** — it carries the largest per-architecture LOAO error (18.3%, reported in §3) — not excluded. All other rows are single-run binary-search estimates.

Model	Arch	Params	tok/call	α_{eff}	α_{exec}	Notes
Pythia-70M†	gpt_neox	70 M	208	1.37	2.15 $\diamond$	Q4_0 SIMD, untied
SmolLM2-135M†	llama	135 M	97	1.53	1.53	Q4_0 SIMD
Gemma3-270M-IT†	gemma3	270 M	56	1.32	1.33	Q4_0 SIMD
OpenELM-270M	openelm	270 M	45	1.65	1.64	Q4_0 SIMD
SmolLM2-360M	llama	360 M	36	1.54	1.54	Q4_0 SIMD
Qwen 2.5 0.5B†	qwen2	494 M	31	1.31	1.31	Q4_0 SIMD
H2O-Danube3-500M	llama	500 M	27	1.48	1.60	Q4_0 SIMD, untied
DistilGPT2	gpt2	82 M	119	2.05 $\triangle$	2.07	legacy regime
GPT-2-124M	gpt2	124 M	84	1.92 $\triangle$	1.93	legacy regime

α_{exec} = *executed-FLOPs convention* (§2): identical to α_{eff} for tied-embedding models; $\diamond$ marks the small-dimension overhead regime ($d \leq 512$) that the published-parameter convention masks. Full recompute incl. the four models not shown: `artifact/results/current/scaling_low/alpha_exec.csv`.

Table 1b. Network and deployment rows measured on ICP mainnet or Swiss Subnet (SSN). These rows are included in the artifact, but they are not all used in the modern-only scaling-law fit above: some are same-build network confirmations, some are build-specific deployment observations, some are systems demonstrations, and some are negative/operational evidence.

Model / setup	Env	Quant / mode	tok/call or outcome	Role in paper
Qwen 2.5 0.5B, our fork	ICP main net	Q8_0	29	Production 2.9× software-gap result
Qwen 2.5 0.5B, onicai baseline	ICP main net	Q8_0	10	Baseline for same canister / weights
SmolLM2-135M	SSN	Q4_0	97	Same-band network validation
Gemma3-270M-IT	SSN	Q4_0	52	Same-band network validation (−7% vs local)
Qwen 2.5 0.5B	SSN	Q4_0	30	Same-band network validation (−3% vs local)
Qwen3.5 0.8B	SSN	Q4_0	13	Highest-quality SSN decoder row; crashes ICP mainnet
Qwen3.5 0.8B	SSN	TQ2_0	18	Throughput valid, quality invalid; negative result
Qwen3.5 0.8B-21L	SSN	Q4_0	18	Pruning throughput valid, quality invalid
Qwen3.5 0.8B-12L	SSN	Q4_0	20	Pruning throughput valid, quality invalid
Qwen3.5 0.8B-12L	SSN	TQ2_0	25	Retracted throughput gain: quality invalid
Qwen3.5 0.8B	SSN	IQ2_XXS	5	Aggressive quantization; not used as evidence
Falcon-H1-Tiny-90M	SSN	Q8_0	199	Build-specific hybrid Mamba+Transformer observation; used for §6.2 IO analysis, not same-build calibration
TriLM 560M	SSN	TQ2_0	34	Ternary floor validation, 3 reps, byte-identical
TriLM 3.9B	SSN	TQ2_0	5	Ternary floor validation, 3 reps, byte-identical
SmolLM2-135M	SSN	BASE-Q8_0	95	Mixed-precision Pareto-winner cross-env validation (§6.1)
Qwen 2.5 0.5B	SSN	BASE-Q8_0	28	Mixed-precision Pareto-winner cross-env validation (§6.1)
Qwen 2.5 0.5B	SSN	BASE-Q8_0, 10-turn	100 total tokens	Multi-call session demo (§6.2)
EmbeddingGemma-30 0M	ICP / SSN	Q4_0 encoder	126 input tok/call	Encoder systems demonstration

The full network tables are `artifact/data/onchain/icp_mainnet.csv` and `artifact/data/onchain/ssn_mainnet.csv`; Paper 1.5 deployment rows are also summarized in `artifact/data/paper_1_5/ternary_measurements.csv`, `mixed_precision_measurements.csv`, and `multicall_characterization.csv`.

Table 1c. Additional Qwen 2.5 0.5B canister characterization. These rows explain the software gap and prompt-length limits; they are not additional same-status calibration anchors.

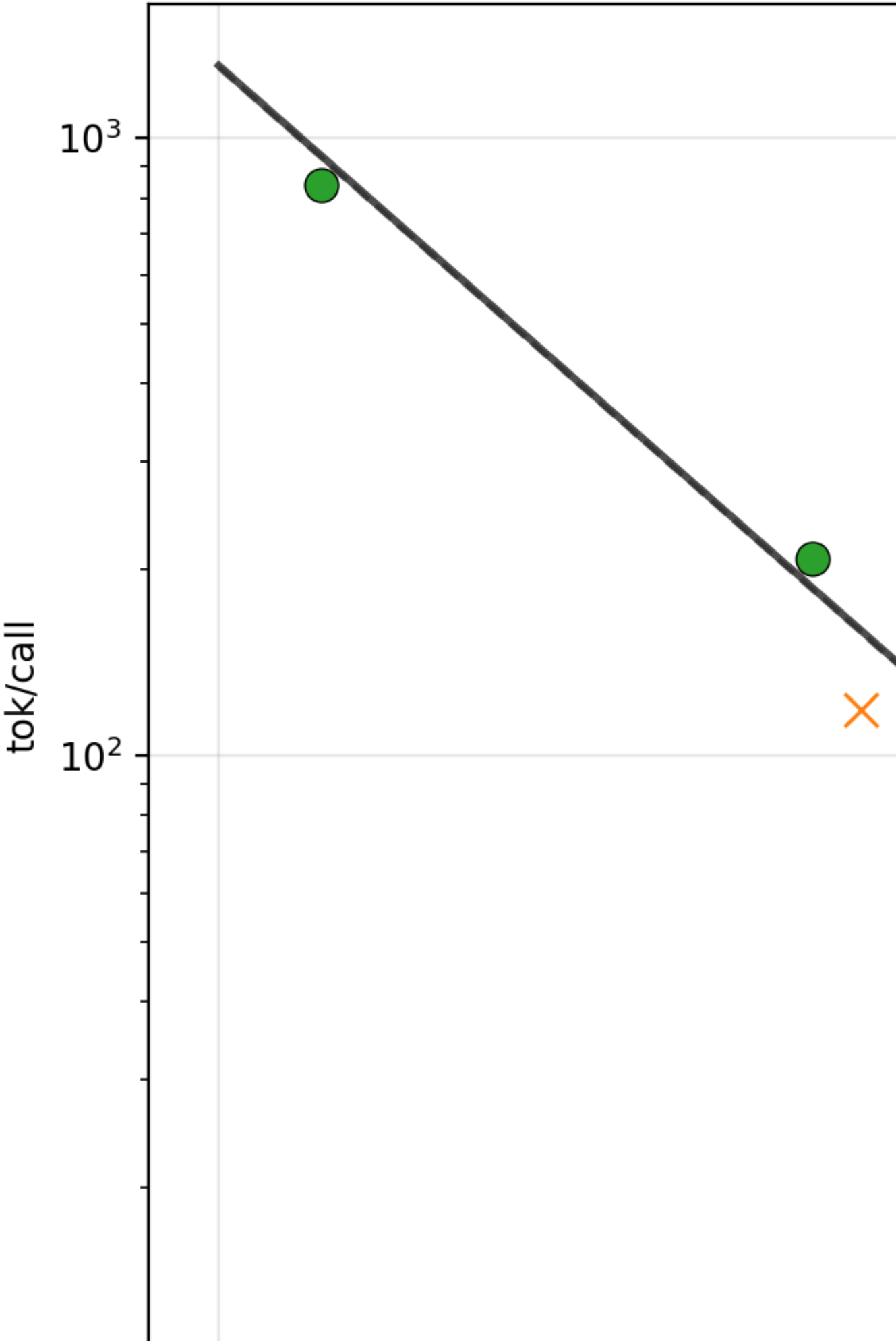
Measurement	Value	Interpretation
Q8_0 SIMD, local canister	30 tok/call	Same optimized path as the 29 tok/call mainnet row, within build/network drift
Q8_0 scalar, local canister	6 tok/call	Scalar path is practically unusable at 0.5B scale
F16, local canister	~3 tok/call	No F16/F32 SIMD path; quantized SIMD is required
Decode profiling, Q8_0 SIMD	12.327B instructions / 10 tokens	$\alpha_{\text{eff}} = 1.25$ in the optimized path (494 M nominal)
MUL_MAT share, same profile	98.3%	The Qwen profile independently confirms matmul dominance
Estimated scalar α_{eff}	~6.75	SIMD compresses effective cost by about 5.5× for Qwen
Prompt ~5 tokens	0 prefill calls; ~28–30 gen tok/call	Short guardrail prompts can stay single-call
Prompt ~50 tokens	2 prefill calls; ~20 gen tok/call	Chat-style prompts pay visible multi-call prefill cost
Prompt ~120 tokens	4 prefill calls; ~10–17 gen tok/call	Long prompts are latency-bound before generation

The underlying registry rows are in `artifact/results/raw/core_measurements_v2.csv`; the network rows are in `artifact/data/onchain/`. The long-form measurement inventory is preserved in `artifact/notes/paper1_long_measurement_inventory.md` for reviewers who want the distinction between fit rows, deployment rows, negative results, and canister-characterization rows.

Because α_{eff} is *defined* as $B / (N \cdot 2P)$, the relation $\text{tok/call} \approx B / (\alpha_{\text{eff}} \cdot 2P)$ is a re-parameterization of the standard 2P FLOP count [16], not an independently discovered law; a log-log slope near -1 is implied by the convention whenever α_{eff} varies far less than P . The falsifiable empirical content is therefore (a) how tightly α_{eff} clusters across heterogeneous families, and (b) whether the other families’ coefficient predicts a held-out architecture’s throughput (LOAO). Fit statistics on the modern-only regime (`artifact/results/current/scaling_law/`, via `artifact/scripts/analyze_scaling_law.py`; fit membership is declared in the registry — `role=calibration` rows with `quant` $\in \{Q4_0, Q8_0\}$ and SIMD enabled, excluding degenerate-output and instruct variants; the quant-homogeneity clause also excludes the sole Q3_K_M row, Bloomz-560M, whose inclusion would move the median from 1.527 to 1.535, +0.5%):

- Median $\alpha_{\text{eff}} = 1.53$, 95% bias-corrected bootstrap CI [1.37, 1.65] (BCa, 9 999 resamples; with $n = 11$ models the median’s bootstrap interval is an order-statistic range whose endpoints coincide with observed models — we report it at two decimals accordingly)
- Strict modern-only LOAO MAPE = 7.7% (model-weighted; family-equal weighting gives 8.9%), max per-architecture error 18.3% — the out-of-sample test that carries the claim
- Log-log slope -0.980 , 95% OLS CI $[-1.047, -0.913]$, R^2 0.9919 — a consistency check on the 2P exponent, not independent validation
- Under the executed-FLOPs convention: α_{exec} median 1.54, CV 9.3% across nine of the eleven models (all-11 CV: 28%); Pythia-70M/14M ($d = 512/128$) sit at 2.15 / 3.12 — an overhead effect at small hidden dimensions that the published-parameter convention masks by crediting never-executed lookup parameters (`alpha_exec.csv`, script `compute_alpha_exec.py`). At $n = 2$ — both models gpt_neox — dimension and family are confounded, and legacy GPT-2 ($d = 768$, $\alpha_{\text{exec}} \approx 2.0$) shows dimension alone is not the controlling variable; we report the split as an observation, not a law. The LOAO test operates on the published-parameter convention — the deployment setting, where only public counts are available — and covers all 11 models.

A. Scaling law with m



Log-log fit of decode throughput against 2P across the modern calibration set (published-parameter convention). The -0.980 slope is a consistency check on the 2P exponent; the informative content is the tightness and transferability of the α_{eff} cluster (§3).

Legacy GPT-2 rows land at $\alpha \approx 2.0$ — systematically above the modern cluster, and only on ICP; on native Apple M4 Max the same models show no regime split (`artifact/data/native/m4_max_baseline.csv`), locating the penalty in the llama.cpp GPT-2 code path on this stack rather than in the architecture. Conventions matter here in an instructive way: GPT-2’s GGUF exports store a duplicated output head (+24–32% audited stored tensors; `discrepancies.csv`), which under a stored-tensor reading would compress α toward the cluster (≈ 1.4) — but duplicated tensors are not executed twice, and under the executed-FLOPs convention GPT-2 remains at $\alpha_{\text{exec}} \approx 1.9\text{--}2.1$ (tied head; learned positions excluded). The legacy penalty is real under the physical convention; we retain GPT-2 as a sensitivity regime, excluded from the modern fit.

Under the published-parameter predictor, architecture details (depth, hidden size, vocabulary) add little once total parameter count is fixed — though not nothing: the depth-pruned AMD-LLaMA-135M runs $\approx 29\%$ faster than same-size SmolLM2 (excluded from the fit for degenerate output; registry row M22), and under α_{exec} the two smallest-dimension models separate (above). Danube3-500M and Qwen 2.5 0.5B occupy very different layer/hidden geometries but share a throughput band because they share a parameter band. Non-Transformer coverage is preliminary: RWKV-7 [12] lands consistent with the cluster; Mamba-370M [13] drifts between builds ($\alpha = 1.54 \rightarrow 1.80$, -14% tok/call) and we treat its result as suggestive, not settled.

Sensitivity to the parameter convention. 2P uses published parameter totals, applied uniformly; the α_{exec} treatment of §2 is the principled resolution of convention questions. Stored-tensor audits are *file-dependent* — the Qwen 2.5 0.5B Q8_0-instruct export stores a duplicated head (630 M stored vs 494 M published) while its Q4_0/TQ2_0 exports store 494 M (`discrepancies.csv`) — but stored duplication does not change execution: **α_{exec} is invariant to it.** The residual convention exposure is therefore confined to α_{eff} ’s role as a *predictor*, where the published count is exactly what a deployer has, and the LOAO error already prices in the untied-model spread.

Absolute tok/call numbers are build-sensitive within $\pm 15\%$ between March and April 2026 builds. The scaling-law *structure* — slope, model ordering, α_{eff} median — is stable across those builds; calibration points must be tied to a concrete WASM hash.

4 Production Validation on ICP Mainnet and Matmul (MUL_MAT) Dominance

4.1 The 2.9 $\times$ fork gap on mainnet

We measured the same Qwen 2.5 0.5B Instruct Q8_0 GGUF (SHA-256 ca59ca7...) on ICP mainnet canister zmm32-7yaa-aaaad-qlsq-cai under the same prompt and the same 40 B per-call instruction cap:

Fork	WASM hash	max gen tok/call	Result
Ours	ef8f9d78...	29	OK
onikai [9]	6c77a958...	10	OK

The 2.9 $\times$ gap is a production result, not a local artefact (`artifact/data/onchain/icp_mainnet.csv`) — a single-run, one-model, one-prompt mainnet measurement (claims-matrix C4) whose deterministic trap-boundary protocol and independent rebuild (below) carry its reliability. Identical weights and identical boundary produce a 2.9 $\times$ throughput difference only because the software path differs.

Independent re-validation (2026-05-19). The functional claim was re-validated on a local dfx replica using freshly rebuilt WASMs from current public sources: Comoto fork at `Simlowker/gian@8cda13b` (rebuild SHA-256 da112d99...) and onikai baseline at `onikai/llama_cpp_canister@v0.10.1` (rebuild SHA-256 b6ccbff0...). Using the identical Qwen 2.5 0.5B Q8_0 GGUF (SHA-256 ca59ca7f...) and identical prompt and protocol, the rebuild yields **Comoto = 29 tok/call, onikai = 10 tok/call, ratio = 2.9 $\times$** — exactly matching the original 2026-04-09 mainnet measurement. The original WASM ef8f9d78... is not byte-reproducible: it was built from working-tree state that was never committed to git (a `.gitignore` pattern shadowed 114 per-model `.cpp` files in `src/models/`; commit 668b48e later imported them as a corrective fix). Byte-exact rebuild is therefore impossible, but functional behavior — the claim itself — is fully reproducible from current public source by any reviewer with dfx, ~ 50 GB disk, and ~ 30 min CPU; no ICP cycles required. The end-to-end procedure is in the companion `instruction-bounded-inference-artifact` repository, file `REPRODUCE.md`; the reproduced 29-vs-10 result is logged in `results/rebench_2026-05-19.md` in that repository.

What the 2.9 $\times$ is made of. The gap is kernel-path vectorization, not hardware or weights: on the same build, the same Qwen Q8_0 runs at 6 tok/call on the scalar path and 30 tok/call with WASM SIMD enabled (Table 1c) — a $\sim 5\times$ spread that brackets the onikai baseline’s 10 tok/call between the scalar and fully-vectorized operating points. The dominant contributors on this stack are SIMD dispatch of the quantized matmul inner loops and the associated dequantization

strategy (§4.2); everything else is bounded by the 1.3–1.7% non-matmul profiling share. We did not characterize the onicai fork’s exact kernel configuration, and the claim is about stack-dependence of α_{eff} , not baseline optimality: an equivalently-vectorized baseline would be expected to close most of the gap.

4.2 Matmul dominates at 98.7%

Per-operation profiling at ggml-operation boundaries on Pythia-70M Q8_0 decode (`artifact/data/profiling/per_operation.csv`) attributes:

Operation class	Share of total cost
MUL_MAT	98.7%
All non-matmul work combined	1.3%

This sharpens the optimization picture. Integer normalization, alternate attention formulations, and other non-matmul tweaks target $\leq 1.3\%$ of cost on the profiled Pythia-70M Q8_0 view ($\leq 1.7\%$ on the Qwen 0.5B Q8_0 SIMD profile reported in §3); residual kernel-table opcode-mix differences ($\approx 4:3$ at the SIMD level — the $2\times f32/i32$ gap is scalar-only, §2) are drowned by metering overhead $\beta = 1.844$ (`artifact/notes/beta_derivation.md`). The only lever that moves end-to-end throughput is effective matmul cost. Matmul-free language modeling [18] is an orthogonal architectural response to the same dominance pattern; for a decoder running in our metering regime, eliminating matmul opcodes outright would change the cost decomposition itself rather than reduce the coefficient.

4.3 Why unstructured sparsity fails (Theorem 1)

Theorem 1. *For any sparsity rate $s \in [0, 1]$, per-element zero-checking strictly increases total instruction cost relative to a sparsity-oblivious matmul on the integer-matmul ICP WebAssembly paths measured here (TQ2_0, Q4_0, Q8_0).*

Proof sketch. The `i32.load` of `w_i` is required by both the baseline MAC and the sparsity check, so it is shared and not counted as overhead; the check inserts `i32.eqz (1) + br_if (2) = 3` additional ICP units per weight. If `w_i = 0`, the check skips a MAC whose cost is 1 unit on the TQ2_0 ternary path (heavier on Q8_0/Q4_0 where the saving stays ≤ 2). Net per zero on TQ2_0: $+3 - 1 = +2$; net per non-zero: $+3$. Total additional cost on the ternary path: $N \cdot (3 - s) > 0$ for all $s \leq 1$. The same strict positivity holds a fortiori on heavier-MAC integer paths where the per-zero saving never matches the $+3$ overhead. The SIMD variant (16-lane zero-check) and the branchless `select` variant both remain strictly positive under analogous counting (`artifact/data/kernel/matmul_bench.csv`). $\square$

Scope. The theorem covers per-element branching, SIMD all-zero checking, and branchless `select` on the integer-matmul paths in this study. It does not apply to *structured* sparsity (N:M patterns with block-level skipping), compressed formats (CSR/CSC), or bitmap indexing that amortizes the check cost. F32-path sparsity is also out of scope: `f32` opcode pricing differs and the per-MAC saving structure changes.

4.4 Cross-subnet consistency

SSN Q4_0 deployments (`artifact/data/onchain/ssn_mainnet.csv`) stay inside the calibration band: SmolLM2-135M matches local at 97 tok/call, Gemma3-270M within 7%, Qwen 2.5 0.5B within 3%. These deltas are consistent with ordinary build and deployment variance, not a different governing law.

5 Reaching the Ternary Instruction-Cost Floor (TQ2_0): $\alpha_{\text{eff}} \approx 1.00 / \alpha_{\text{exec}} \approx 1.17$

5.1 TriLM ternary: $\alpha_{\text{eff}} = 1.03$ at 560 M, consistent at 3.9 B

The floor evidence in this section comes from one model family (SpectraSuite TriLM [15]) and one format (TQ2_0); within that scope it is measured at two scales $7\times$ apart. We integrated a custom TQ2_0 (2-bit ternary, weights in $\{-1, 0, +1\}$) WASM SIMD path — absent from upstream `llama.cpp` [10] as of this writing — end-to-end into the inference framework, and measured two purpose-trained ternary models on the same canister build (`artifact/data/paper_1_5/ternary_measurements.csv`, `artifact/results/paper_1_5/PHASE-1-SUMMARY.md`). The broader ternary-LLM line — BitNet-style 1-bit and 1.58-bit training [3, 22], ternary scaling laws and efficient inference on the same TriLM suite [31], ternary sparsification [17], edge-inference kernels [4], CPU SIMD reorganizations for ternary inference [27], and recent multiplication-free fused kernels and consumer-CPU ternary SIMD frameworks [32, 33] — motivates the integer-matmul boundary we exploit here; we differ in targeting a consensus-metered WASM canister rather than native hardware:

Metric	TriLM 560M	TriLM 3.9B
Params (GGUF-audited)	569 M	3 992 M
GGUF size (TQ2_0)	195 MB	1 113 MB
Architecture	24 L $\times$ 1280 H	30 L $\times$ 3072 H

Metric	TriLM 560M	TriLM 3.9B
N_MAX local dfx (3 reps)	34 (CV 0%)	5 (CV 0%)
N_MAX SSN mainnet (3 reps)	34 (CV 0%)	5 (CV 0%)
N_MAX ICP mainnet, standard subnet (3 reps, 2026-07)	34 (CV 0%)	—
Implied α_{eff} (published-param)	1.03	1.00
Implied α_{exec} (executed FLOPs, §2)	1.17	≤ 1.04 (one-sided)
Modern-regime baseline ($\alpha = 1.54$, §3)	22.8	3.3
Speedup vs baseline	1.49×	$\approx 1.5\times$
Coherence	10/10 fluency	factual ✓ at N=5

The 560 M point is the tight anchor: $N_{\text{MAX}} = 34$ implies $\alpha_{\text{eff}} \in (1.004, 1.034]$ — within 3% of the floor. The 3.9 B point is a coarse consistency check, not independent floor evidence: at $N_{\text{MAX}} = 5$ the integer-token resolution is asymmetric ($-17\%/+0.2\%$), so the observation establishes $\alpha_{\text{eff}} \leq 1.002$ *given* the decode-path floor $\alpha \geq 1$, and distinguishes little below it. We therefore read the pair as evidence consistent with the floor on the measured TQ2_0 path, not as closed saturation. Under the executed-FLOPs convention the same measurements read $\alpha_{\text{exec}} = 1.17$ at 560 M and ≤ 1.04 at 3.9 B: the ternary path sits far below the modern cluster (≈ 1.54) under both conventions, and the decomposition becomes transparent — ≈ 0.5 units/FLOP of ternary arithmetic (Theorem 1 accounting) plus unamortized per-weight load/dequant. The proximity of the published-parameter values to exactly 1.0 is partly a coincidence of TriLM’s untied input table (§2). The $\approx 1.5\times$ speedup over the modern-regime baseline is constant across the $7\times$ span, weakening the alternative explanation that the 560 M result was inflated by small-scale inefficiency in the baseline.

5.2 Byte-exact cross-environment determinism

A stronger consistency check than throughput matching is bit-for-bit output agreement — and we now have it across **three execution environments**. In April 2026, the 10-prompt TriLM 560M coherence audit on the 13-node SSN canister diffed empty against the local dfx run (`artifact/results/paper_1_5/raw/trilm-560m-ssn-vs-mainnet-diff.txt`), and for TriLM 3.9B every token produced at $N = 1..5$ on SSN matched the local output verbatim. In July 2026 we repeated the campaign on a **standard ICP mainnet application subnet** (canister `zmm32-7yaaa-aaaad-qlsq-qai`, subnet `5kdm2-...`), this time with both sides of every diff archived (`artifact/results/paper_1_5/raw/mainnet-2026-07-09/`): $N_{\text{MAX}} = 34$ in 3/3 repetitions with the boundary trap at 35, and the 10-prompt audit **byte-identical between the local replica and the mainnet canister** — under an independently rebuilt GGUF (a separate conversion pipeline, three months after the original) and a newer canister build, with all completions word-for-word equal to the April audit. Within a single subnet, replica agreement is protocol-guaranteed — a divergent replica would break consensus — so the novel content here is the byte-equality *across environments, build generations, and weight-conversion pipelines*, which no protocol layer guarantees; that equality is inconsistent with non-deterministic kernel hazards (uninitialized vector lanes, FP reorder, replica-specific code paths) on the measured paths. The TQ2_0 WASM SIMD kernel is fully deterministic under live consensus on both production subnets.

5.3 Throughput density

On non-purpose-trained ternary weights, the same TQ2_0 path reaches **201 tok/call on Pythia-70M**, giving **5.03 tok/MB** — the highest throughput-per-megabyte observed in the study (`artifact/results/current/extended_analysis/`). Note what this row does *not* show: at 201 tok/call, Pythia-70M TQ2_0 runs at $\alpha_{\text{eff}} \approx 1.41$ ($\alpha_{\text{exec}} \approx 2.2$, consistent with its small-dimension regime, §3) — well above the floor, and marginally below its own Q4_0 SIMD throughput (208 tok/call, Table 1). The floor of §5.1 is attained by purpose-trained ternary models at sufficient scale on this path, not by the TQ2_0 format per se. Q4_0 SIMD remains the best absolute-throughput operating point for non-ternary models on the calibration build (Table 1, Q4_0 31 vs Q8_0 30 for Qwen); the ordering is build-sensitive — §6.1’s campaign build showed Q8_0 ahead, its Q4_0 value being the documented build regression — while §6.1’s Pareto conclusion (uniform Q8_0 non-dominated) is robust to that drift. A ternary-native DOT $\times 4$ microkernel projects a $2.45\times$ bench-level speedup over F32 SIMD (artifact only; no model-level run yet).

6 Pareto Quantization (Mixed-Precision vs Uniform Q8_0) and the Multi-Call Session IO Tax

6.1 Q8_0 stays on the Pareto frontier

The per-call law does not by itself say which quantization to pick. We computed per-layer ΔPPL sensitivity on WikiText-2, generated six mixed-precision variants per model (embedding $\rightarrow$ Q8, attention $\rightarrow$ Q8, first K layers $\rightarrow$ Q8, top- ΔPPL layers $\rightarrow$ Q8) on a Q4_K_M base, and compared against uniform Q4_0, Q5_K_M, Q8_0 on SmolLM2-135M and Qwen 2.5 0.5B. Throughput rows in `artifact/data/paper_1_5/mixed_precision_measurements.csv`; perplexity rows in `artifact/data/paper_1_5/c2-ppl-smolllm135-only.csv` (SmolLM2) and `artifact/results/paper_1_5/raw/c2-ppl-qwen05-only.csv` (Qwen); joined summary in `artifact/results/paper_1_5/tables/c2-mixed-precision-summary.md`.

Table 2. Pareto winners on both models. Qwen 0.5B Q8_0 size is reported as the Phase 2 local build (506.5 MB, matching the anchored throughput row); the Phase 3 SSN rebuild is 531 MB with byte-exact weights, differing only in metadata.

Metric	SmolLM2-135M	Qwen 2.5 0.5B
	BASE-Q8_0	BASE-Q8_0
Preferred frontier point		
size (MB)	144.8	506.5
tok/call	97	30
PPL WikiText-2	13.603	11.680
PPL WT2 (F16 baseline)	13.574	11.674
Δ PPL vs F16	+0.029	+0.007
$\alpha_{\text{eff}} = B / (N \cdot 2P)$	1.527	1.333

α_{eff} in this table uses rounded nominal parameter counts (135 M / 0.5 B), giving 1.527 / 1.333; on the registry's nominal counts (134.5 M / 494 M) the values are 1.53 / 1.35 — the same convention as Table 1. The Qwen Q8_0-instruct GGUF export audits at 630 M tensor-level parameters (discrepancies.csv), on which its α_{eff} would be ≈ 1.06 ; see the parameter-convention sensitivity note in §3.

Uniform Q8_0 is the highest-throughput, lowest-PPL point on both tested frontiers, but not the smallest. The precise statement is: no mixed variant strictly dominates it. Q5_K_M is the sole strictly-dominated variant. The Δ PPL-guided V4 variant (top-K fragility layers at Q8, rest at Q4_K_M) is the best mixed variant and remains relevant under tight size budgets.

Three caveats. First, neither hidden dim (576, 896) is divisible by 256, so llama-quantize emits many q4_K/q5_K $\rightarrow$ q5_0/q8_0 fallbacks; the mixed-precision landscape likely differs on hidden dims in {512, 1024, 2048}. Second, Qwen's $\alpha_{\text{eff}} = 1.33$ is lower than SmolLM's 1.53 despite larger hidden dim. Plausible contributors: GQA (grouped-query attention; fewer K/V FLOPs), attention biases not counted in 2P, and SmolLM2's tied embedding making 2P undercount its LM head. Both values remain inside the modern-regime CI. Third, cross-campaign build drift: this campaign's absolute BASE-Q4_0 throughput for Qwen (20 tok/call on the campaign build) sits 35% below the Table 1 calibration value (31) — a model-specific regression of that build line (SmolLM2 moved only 97 $\rightarrow$ 95 on the same build), re-measured at **30 tok/call** on the current release build (−3%, inside the $\pm 15\%$ envelope; probes and provenance checks in artifact/results/paper_1_5/raw/qwen-q4-build-drift-2026-07-09.md). Within-campaign, same-build comparisons — the Pareto analysis above — are unaffected; absolute values should always be read against the pinned WASM hash (Limitation 4).

Validated on SSN. Pareto-winner BASE-Q8_0 on SSN mainnet: SmolLM2-135M N_MAX = 95 (local = 97, −2%, byte-exact 3/3 reps); Qwen 2.5 0.5B N_MAX = 28 (local = 30, −6.7%, byte-exact 3/3 reps). Source: artifact/results/paper_1_5/tables/c2-c3-ssn-validation-summary.md.

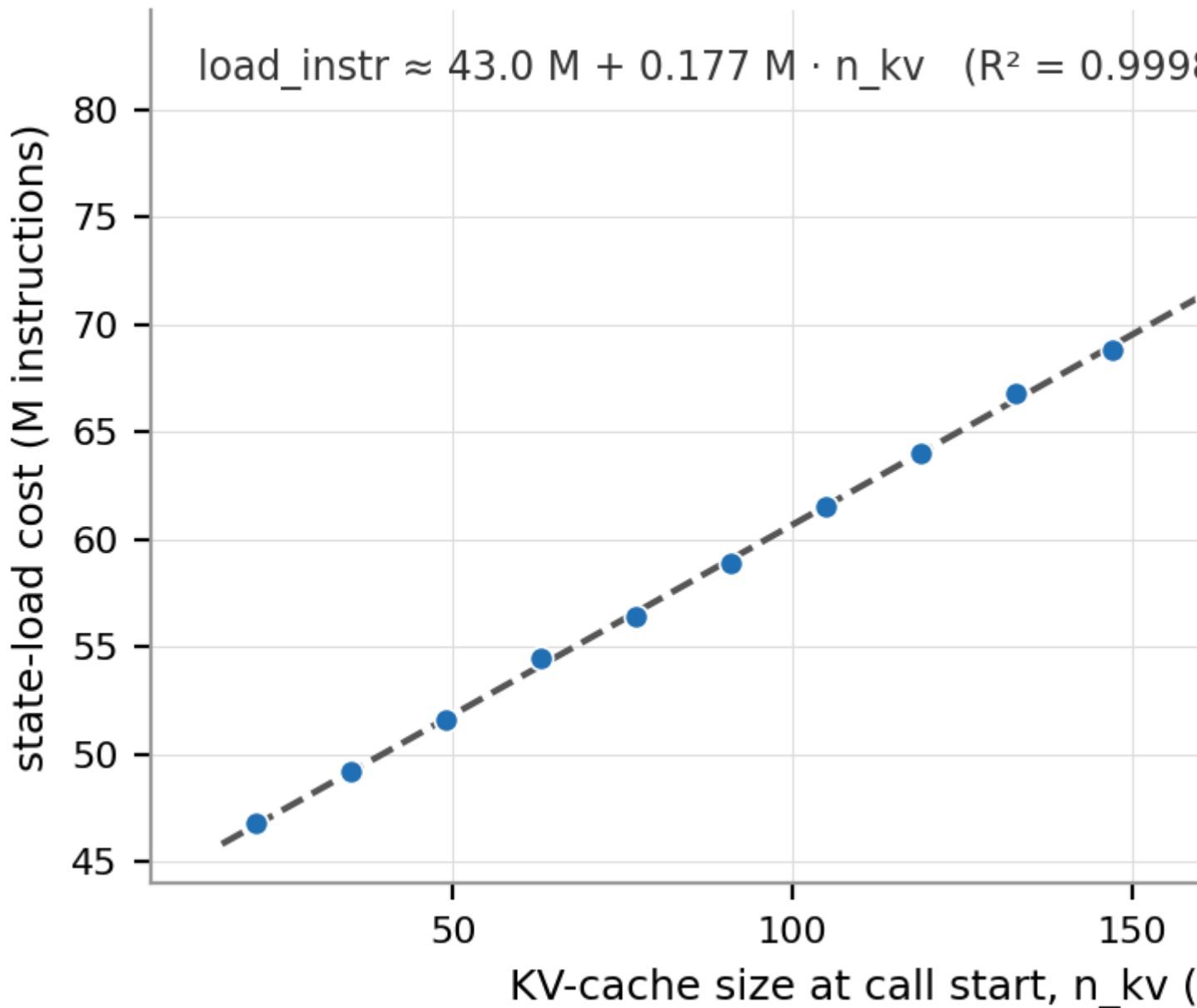
6.2 Multi-call stateful session budget

A single 40 B call is often not enough for agentic workloads. Using --prompt-cache-all, a session of N calls pays two extra per-call IO costs the single-call formula ignores: save_instr (serialize the growing KV cache at end-of-call) and load_instr (deserialize at start of calls 2..N), both in raw instructions. These reduce the effective per-call decode budget, giving an **extended session formula**:

$$\text{tok/call}_{\text{effective}} \approx (B - \text{IO_instr}(n_{\text{kv}}, \text{cache_type})) / (\alpha_{\text{eff}} \times 2P)$$

$$\text{tok/session} \approx N \times \text{tok/call}_{\text{effective}}$$

with $\text{IO_instr}(n, c) = \text{save_instr}(n, c) + \text{load_instr}(n, c) \approx (a_s + b_s \cdot n) + (a_l + b_l \cdot n)$ fitted from instrumented canister runs (an instrumented build — ggml commit bb238b9 — logs save_instr and load_instr per call). Table 3's campaign fit resolved the Qwen load cost to a near-constant ($b_l \approx 0$ over that build's operating range); a dedicated 15-point sweep on the current release build (one growing session, n_{kv} 21–217) measures the load slope directly: $\text{load_instr} \approx 43.0 \text{ M} + 0.177 \text{ M} \cdot n_{\text{kv}}$, $R^2 = 0.9998$ (figure below; artifact/results/paper_1_5/raw/bl-sweep-2026-07-09/). The load slope is real but small — $\approx 19\%$ of the save slope — so the near-constant reading holds at small caches, while the growth term belongs in long-session budgets. Absolute IO constants are build-specific (the current build loads $\approx 10\times$ cheaper than Table 3's campaign build at comparable cache sizes); the linear *structure* is the transferable finding. The near-unity R^2 reflects a deterministic metered runtime over evenly-spaced points from one session, one model, and one build — it certifies linearity over the measured range (n_{kv} 21–217), not cross-model generality, and beyond that range the formula is extrapolation. At measured cache sizes the two-term tax is a low single-digit percentage of B; its significance for session budgeting is predictability, not magnitude.



Units: B and IO_instr in ICP instruction-cost units; $\alpha_{\text{eff}} \times 2P$ in instructions per token.

Table 3. Fitted IO parameters (artifact/data/paper_1_5/multicall_characterization.csv, artifact/results/paper_1_5/tables/c3-fit-parameters.json).

Model / cache (f16)	a_s (M)	b_s (M/tok)	R²	load mean (M)	load CV
Qwen 2.5 0.5B Q4_0 (Transformer)	26.7	0.951	0.998	425.6	0.02%
Falcon-H1-Tiny-90M Q8_0 (hybrid Mamba+T)	276.3	0.519	0.972	108.3	3%

Observations. The hybrid Mamba+Transformer baseline (Falcon-H1-Tiny, in the broader family of hybrid-head SSM/attention small models [26]) halves the per-token save slope (0.52 M vs 0.95 M) and cuts the load cost $\approx 4\times$ at campaign cache sizes (108 M vs 426 M means on that build) — and a revision-round refit shows its load, too, grows linearly rather than sitting at a constant: ≈ 0.166 M per cached token across the campaign’s four steps ($R^2 > 0.999$, slope from consecutive differences), of the same order as the Qwen current-build sweep slope (0.177). Neither model eliminates per-token growth — attention blocks still accumulate KV. The SSM advantage is partial, not the full “bounded state” headline. `--cache-type-k {f16, q8_0, q4_0}` is accepted at `load_model` but is a no-op on the current WASM (KV allocated before the flag reaches the cache-swap path); a rebuild exposing it at runtime is the obvious next lever. Re-attach latency is dominated by `load_model` itself (~ 2.3 s wall-clock), not by state IO (artifact/results/paper_1_5/raw/c3-reattach-qwen05.log). A 10-turn chat demo on SSN (Qwen 2.5 0.5B Q8_0) completed 10/10 turns, 100 cumulative tokens, 41.12 s wall-clock, with narrative coherence preserved under 13-node consensus (c3-ssn-chat-demo-transcript.md).

For short-turn agents (cache $\ll$ 100 tok) the constant term dominates and Falcon-H1 wins the cold path by $\sim 15\%$; for medium sessions (cache ~ 100 tok) by $\sim 20\%$; for long chat (cache $\gg 500$ tok) the slope difference becomes load-bearing. The extended session formula is the operative budget for any on-canister agentic workload.

6.3 Operational ceilings

Two ceilings bound the deployable envelope. First, on the tested SSN wasm32 build we observed a **practical wasm-heap ceiling near 900 MB**; in the Qwen 0.5B Q8_0 load path, reducing to `-c 512 -b 512` (context and batch) brought the allocation under that envelope and loaded cleanly. The background for non-ICP readers: a wasm32 canister heap is bounded by a 32-bit WebAssembly address space (theoretical 4 GiB), while stable memory is a separate persistent store and current ICP documentation also distinguishes wasm64 limits. The observed ceiling therefore reflects current heap-side allocations — weights staging, KV cache, compute buffers, framework state, metering instrumentation, and scratch space — rather than a single pool shared with stable memory. Treat 900 MB as a build-specific operational observation, not a universal threshold. Second, an IC0524 **page-access trap** triggers on some ≥ 1 GB GGUFs during `load_model`, but not on raw size alone: TriLM 3.9B TQ2_0 at 1 113 MB loads cleanly, while Qwen 2.5 1.5B Q6_K at 1 464 MB traps IC0524 (`artifact/data/paper_1_5/ic0524_characterization.csv`). The failure is format-specific — ternary TQ2_0 is unusually friendly to the ICP page-access pattern. A parallel IC0502 "unreachable" trap at different sizes (Qwen 1.5B Q4_0 at 892 MB) is driven by Wasm-runtime limits rather than stable-memory limits; the two failure modes should not be conflated.

7 Secondary Results (On-Chain Encoder Case Study) and Limitations

7.1 On-chain encoder case study

EmbeddingGemma-300M Q4_0 [11] runs end-to-end inside an SSN canister (`u2pva-3iaaa-aaaax-qaa7a-cai`, 212 MB on disk) at about **126 input tokens per call**, corresponding to $\alpha_{\text{embed}} \approx 0.53$ under the published-parameter convention (`artifact/data/onchain/ssn_mainnet.csv` embedding rows). The apparent sub-floor coefficient is a convention artifact: 201.3 M of the model's 308 M published parameters are an embedding lookup table that executes no FLOPs in the forward pass; under the executed-FLOPs convention $\alpha_{\text{exec}} \approx 1.61$, inside the modern cluster (§2; `artifact/notes/alpha_exec_convention.md`). The per-input-token cost advantage over autoregressive decode is nonetheless real — batched encode amortizes per-weight loads that single-token decode cannot (§2) — so retrieval-style workloads fit the instruction-bounded regime better than long-form chat generation. The deployment is a systems demonstration (N = 1 model, small cross-lingual sanity checks). It is not a claim of priority on the broader "on-chain embedding" term: ecosystem projects under different system boundaries — stored precomputed vectors, off-chain workers, oracle submissions — have existed for some time.

7.2 Minor systems contribution

A WASM SIMD vectorized unpack for the Q6_K q1/qh path (reviewer-flagged scalar path in upstream PR #11453) gives +2.3% `matmul-bench` speedup and 3.2 $\times$ variance reduction (CV 8.0% $\rightarrow$ 2.5%, bit-exact vs the scalar reference). End-to-end on canister the gain stays below integer-token granularity and is kept in the artifact for future Q6_K workloads that might land above the threshold.

7.3 Limitations

1. **Calibration coverage.** The modern-regime law ($\alpha_{\text{eff}} \approx 1.53$) is anchored on the 70 M–800 M range; the ternary floor ($\alpha_{\text{eff}} \approx 1.00$) is validated at 560 M (local, SSN, and a standard ICP mainnet subnet) and 3.9 B (local, SSN) but only on the TriLM family and only for TQ2_0. Neither result is yet a universal law across architecture, quantization, and scale.
2. **Variance coverage.** 4 of 17 modern-Transformer calibration rows are variance-verified (3 reps, CV < 1.05%); the 17 counts `role=calibration` registry rows with modern Transformer architectures in `artifact/data/models.csv`, quantization/SIMD variants counted as separate rows (GPT-2 legacy rows and the Mamba/RWKV non-Transformer baselines excluded) — Pythia-70M Q4_0, SmolLM2-135M Q4_0, Gemma3-270M-IT Q4_0, Qwen 2.5 0.5B Q4_0; a fifth variance-verified point, Mamba-370M, is the non-Transformer baseline included in the modern-only LOAO fit, where it is the single largest per-architecture error (18.3%). The remaining single-run binary-search estimates still contribute to the LOAO summary and the reported modern-only BCa interval in §3, so the interval should be read as conditional on a mixed repeated/single-run evidence base rather than as a repetition-only uncertainty estimate.
3. **Non-Transformer evidence is preliminary.** RWKV-7 is consistent; Mamba is build-sensitive. Cross-architecture generalization is not settled.
4. **Build sensitivity.** Absolute tok/call typically drifts within $\pm 15\%$ between builds — but not always: one build line carried a model-specific -35% regression (Qwen Q4_0 31 $\rightarrow$ 20 tok/call, §6.1; `artifact/results/`

- paper_1_5/raw/qwen-q4-build-drift-2026-07-09.md) that only cross-build re-measurement exposed. Calibration points must be tied to a WASM hash, and per-hash re-measurement is the only reliable guard.
5. **Prefill.** Long-prompt usability remains poor under a per-call budget: a 512-token SmoLLM2-135M prompt requires ~15 sequential calls (~15–30 s latency) before the first generated token. Decode may obey a simple law while user-visible latency still needs multi-call orchestration.
 6. **Throughput is not quality.** Higher tok/call does not imply a better model; it implies a cheaper one under the budget. We retain Qwen 3.5 ternary and layer-pruned variants in the artifact as explicit negative results: throughput-valid but quality-invalid (artifact/data/quality_audit.csv, flagged role=negative_result).
 7. **Embedding evaluation.** The encoder case study is N = 1 with small sanity checks; MTEB-grade on-chain evaluation is a template (artifact/scripts/run_mteb_fra.py), not a completed benchmark.

8 Conclusion

Instruction-metered execution creates a distinct inference regime. On ICP, decode throughput is not best understood through bandwidth language; it is a budgeted compute problem with a hard per-call cap and a stack-dependent effective cost coefficient.

Within that regime we establish three facts on the current release line. A simple first-order law $\text{tok/call} \approx B / (\alpha_{\text{eff}} \times 2P)$ holds for modern decoder families, with a separate penalty regime for legacy GPT-2 code paths. The production explanation is software: the fork gap reproduces on mainnet, matmul is the only first-order optimization lever, and per-element sparsity is formally excluded. Evidence consistent with the empirical decode-path floor is observed on a purpose-built kernel and purpose-trained ternary weights at both 560 M and 3.9 B, under live consensus on both a Swiss-Subnet and a standard-mainnet deployment, with byte-identical outputs across three execution environments — a systems result, not a universal theorem of saturation ($\alpha_{\text{exec}} = 1.17$ under the executed-FLOPs convention, §2; the isolated kernel bench at $\alpha_{\text{kernel}} = 0.817$ shows a further floor below the current decode path, not yet integrated at model level).

Two derived results matter for deployment. Uniform Q8_0 remains non-dominated against the mixed-precision variants tested, and multi-call stateful inference pays a predictable two-term IO tax per call. The practical consequence is narrow: choose architecture for quality; choose parameter count and quantization from the law; treat matmul-path engineering as the primary throughput lever; and budget IO overhead once sessions exceed a single call. The 40 B instruction budget is not the operative deployment ceiling at current scales — wasm heap and page-access limits are. Extensions to larger scales and to ternary-native DOT ×4 at model level are straightforward follow-ups, already scoped in the artifact scripts.

Data and Code Availability

All code, data, rebuild scripts, and raw logs behind every load-bearing claim are released as a companion artifact: code under the MIT License, data under CC BY 4.0. The claim-to-evidence mapping is CLAIMS-EVIDENCE-MATRIX.md; the readiness check artifact/scripts/check_paper_readiness.py validates registry/table/claim consistency.

- **Companion artifact** (calibration registry, per-operation profiling, Paper 1.5 CSVs, scripts, exports): the artifact / directory of this package.
- **Reproduction repository:** github.com/Simlowker/instruction-bounded-inference-artifact — the end-to-end local rebuild of the 2.9× fork gap (REPRODUCE.md) and its result log (results/rebench_2026-05-19.md). Public.
- **Archived snapshot + DOI:** Zenodo 10.5281/zenodo.20607598 (reserved; the record archives the manuscript PDF and the artifact snapshot).
- **Live deployments:** ICP mainnet decode canister zmm32-7yaaa-aaaad-qlsq-qai (re-provisioned 2026-07-09, running the TriLM 560M TQ2_0 model of §5); SSN encoder canister u2pva-3iaaa-aaaax-qaa7a-qai. Canister liveness is cycles-dependent and periodic; the archived logs, not the live canisters, are the durable evidence.

Table DA1. Key artifacts → location → SHA-256 → claim supported.

Artifact	Location	SHA-256 (abridged)	Supports
Qwen 2.5 0.5B Q8_0 GGUF	instruction-bounded-inference-artifact/REPRODUCE.md	ca59ca7f...6844e	§4.1 mainnet 29-vs-10
Comoto fork WASM (rebuilt, Simlowker/gian@8cda13b)	results/rebench_2026-05-19.md	da112d99...a49c	§4.1 (29 tok/call)
onikai WASM (rebuilt, onikai/llama_cpp_canister@v0.10.1)	results/rebench_2026-05-19.md	b6ccbff0...9660	§4.1 (10 tok/call)
Calibration registry (49 rows)	artifact/data/models.csv, artifact/data/onchain/	—	§3 scaling law
Scaling-law outputs (BCa, LOAO)	artifact/results/current/scaling_law/	—	§3 ($\alpha_{\text{eff}} = 1.527$)

Artifact	Location	SHA-256 (abridged)	Supports
Ternary floor (TriLM 560M/3.9B)	<code>artifact/data/paper_1_5/ternary_measurements.csv</code>	—	§5 ($\alpha_{\text{eff}} \approx 1$)
Mixed-precision sweep	<code>artifact/data/paper_1_5/mixed_precision_measurement_s.csv</code>	—	§6.1 Pareto
Multi-call IO fits	<code>artifact/data/paper_1_5/multicall_characterization.csv</code>	—	§6.2 IO tax
Operational ceilings (IC0524)	<code>artifact/data/paper_1_5/ic0524_characterization.csv</code>	—	§6.3

Full SHA-256 values, rebuild commits, baseline tags, and the non-reproducible original-WASM caveat are in `artifact/notes/source_pinning.md`.

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